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 Studierichting: Informatica
 Oefeningenreeks 5B nr. 5.5

$$\begin{aligned}
 & \int_0^1 x^2 \ln x dx \\
 &= \lim_{a \rightarrow 0^+} \int_a^1 x^2 \ln x dx \\
 & \left\{ \begin{array}{l} u = \ln x \\ dv = x^2 dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = \frac{1}{x} dx \\ v = \frac{x^3}{3} \end{array} \right. \\
 &= \lim_{a \rightarrow 0^+} \ln x \frac{x^3}{3} - \int \frac{x^3}{3} \cdot \frac{1}{x} dx \\
 &= \lim_{a \rightarrow 0^+} \ln x \frac{x^3}{3} - \frac{1}{3} \int x^2 dx \\
 &= \lim_{a \rightarrow 0^+} \ln x \frac{x^3}{3} - \frac{1}{3} \cdot \frac{x^3}{3} + C \\
 &= \lim_{a \rightarrow 0^+} \left[\ln x \frac{x^3}{3} - \frac{x^3}{9} \right]_a^1 \\
 &= \lim_{a \rightarrow 0^+} \left(\ln(1) \frac{1^3}{3} - \frac{1^3}{9} \right) - \left(\ln(a) \frac{a^3}{3} - \frac{a^3}{9} \right) \\
 &= \lim_{a \rightarrow 0^+} \left(-\frac{1}{9} - \left(\ln(a) \frac{a^3}{3} - \frac{a^3}{9} \right) \right) \\
 & \left\{ \begin{array}{l} \lim_{a \rightarrow 0^+} a^3 = 0 \rightarrow \frac{a^3}{9} = 0 \\ \lim_{a \rightarrow 0^+} a^3 \ln(a) = 0 \cdot (-\infty) \text{ onbepaald} \\ \Rightarrow \frac{\ln(a)}{\frac{1}{a^3}} = \frac{-\infty}{+\infty} \stackrel{\text{L'Hôpital}}{=} \lim_{a \rightarrow 0^+} \frac{\frac{1}{a}}{\frac{3}{-a^4}} = \lim_{a \rightarrow 0^+} \frac{1}{a} \cdot \frac{a^4}{-3} = \lim_{a \rightarrow 0^+} \frac{-a^3}{3} = 0 \end{array} \right. \\
 &= -\frac{1}{9} - (0 \cdot 0) = -\frac{1}{9}
 \end{aligned}$$

References